Perpendicular bisectors

1

a $y = -2x + 2$

b $x + 2y + 7 = 0$

c $x + 4y - 29 = 0$

d $4x - 3y + 11 = 0$

2

a

i $(3, -1)$ **ii** $m = -1$ **iii** $y = x - 4$

b

i $(-1, -2)$

ii $m = -\frac{2}{3}$

iii $3x - 2y - 1 = 0$

c

i $(0.5, 0.5)$

ii $m = \frac{3}{7}$

iii $7x + 3y - 5 = 0$

d

i $(-1, 1)$

ii $m = \frac{4}{7}$

iii $7x + 4y + 3 = 0$

3

a $x + 7y - 9 = 0$

b $x + 4y - 17 = 0$

c $x - 5y + 60 = 0$

d $5x + 6y - 79 = 0$

e $3x + 7y - 14 = 0$

f $3x + y - 2 = 0$

g $5x - 2y - 21 = 0$

h $x + 2y - 100 = 0$

4 $5x - 2y + 9 = 0$

Applications

5

a $y = x$

b Yes, both points lie on the perpendicular bisector.

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6

a

i $2x + 3y - 39 = 0$

ii $3x - 5y + 2 = 0$

iii $5x - 2y - 37 = 0$

b $(19.9, 6.37)$

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